TDVI: Inteligencia Artificial

Arquitecturas famosas y mapas de activación

Licenciatura en Tecnología Digital

Agenda

1. **Clasificación de imágenes e ImageNet challenge**
2. LeNet
3. AlexNet
4. VGG
5. GoogLeNet + inception
6. ResNet
7. ZFNet
8. Cam
9. Grad-Cam
10. Bibliografía

Clasificación de imágenes

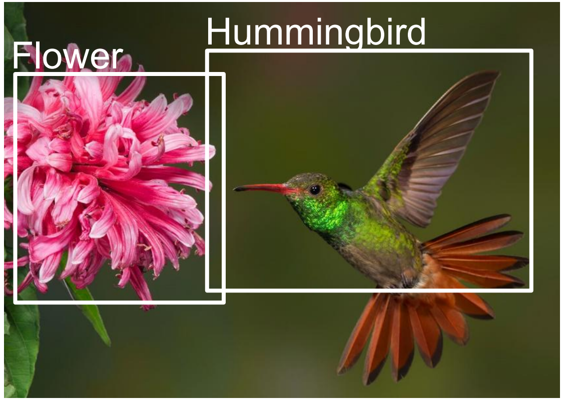
- Las arquitecturas de redes convolucionales CNN fueron diseñadas para entrenar grandes colecciones de datos. (i.e., ImageNet).
- Implementaciones de varios modelos y sus pesos (pre-entrenados) están disponibles en varios frameworks.

Reconocimiento de objetos

Tres tipos de algoritmos para reconocimiento de objetos:

Clasificación de imágenes

Clasificación con localización



Detección

ImageNet: A Large-Scale Hierarchical Image Database

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Abstract

The explosion of image data on the Internet has the potential to foster more sophisticated and robust models and algorithms to index, retrieve, organize and interact with images and multimedia data. But exactly how such data can be harnessed and organized remains a critical problem. We content-based image search and image understanding algorithms, as well as for providing critical training and benchmarking data for such algorithms.

ImageNet uses the hierarchical structure of WordNet [9]. Each meaningful concept in WordNet, possibly described by multiple words or word phrases, is called a "synonym set" or "synset". There are around 80,000 noun synsets

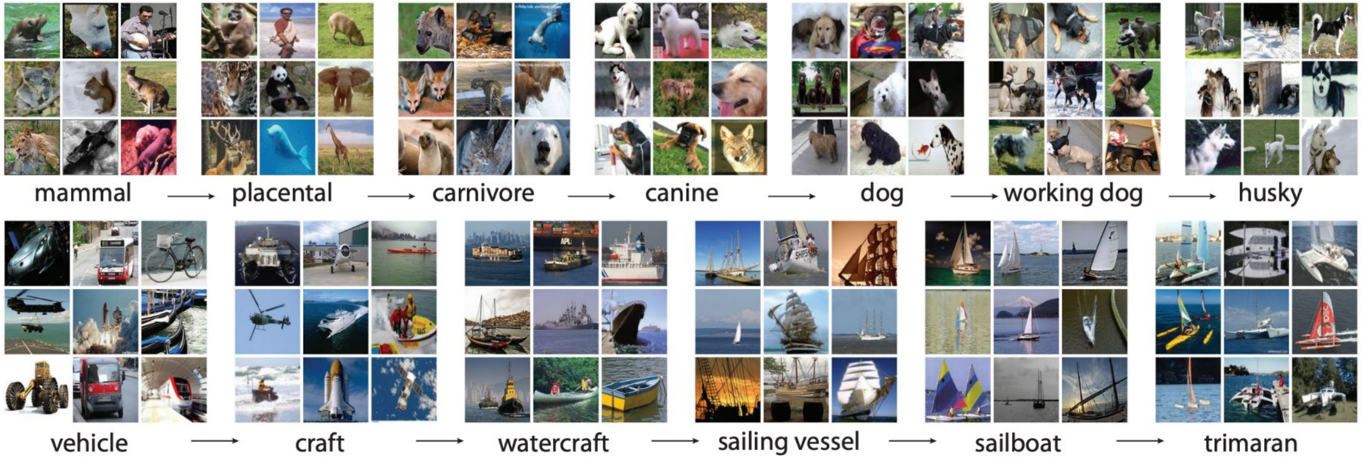
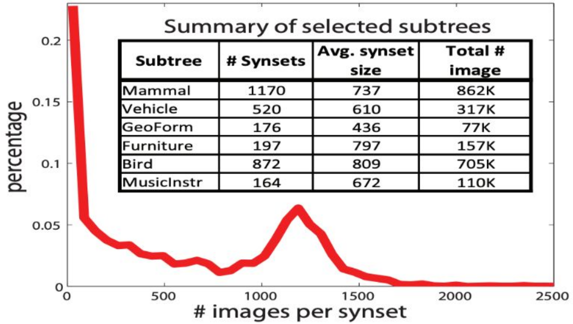


Figure 1: A snapshot of two root-to-leaf branches of ImageNet: the **top** row is from the mammal subtree; the **bottom** row is from the vehicle subtree. For each synset, 9 randomly sampled images are presented.



Subtree	# Synsets	Avg. synset size	Total # image
Mammal	1170	737	862K
Vehicle	520	610	317K
GeoForm	176	436	77K
Furniture	197	797	157K
Bird	872	809	705K
MusicInstr	164	672	110K

Figure 2: Scale of ImageNet. **Red curve**: Histogram of number of images per synset. About 20% of the synsets have very few images. Over 50% synsets have more than 500 images. **Table**: Summary of selected subtrees. For complete and up-to-date statis-

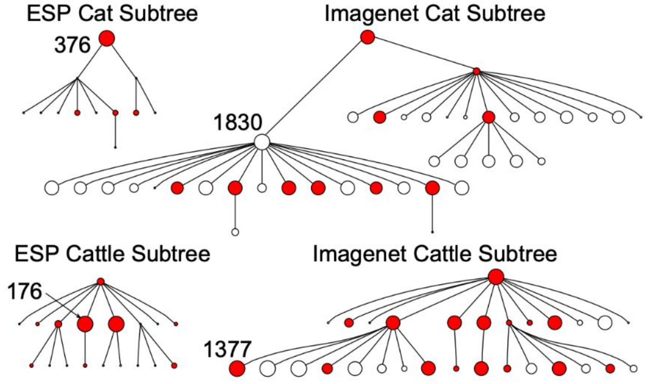
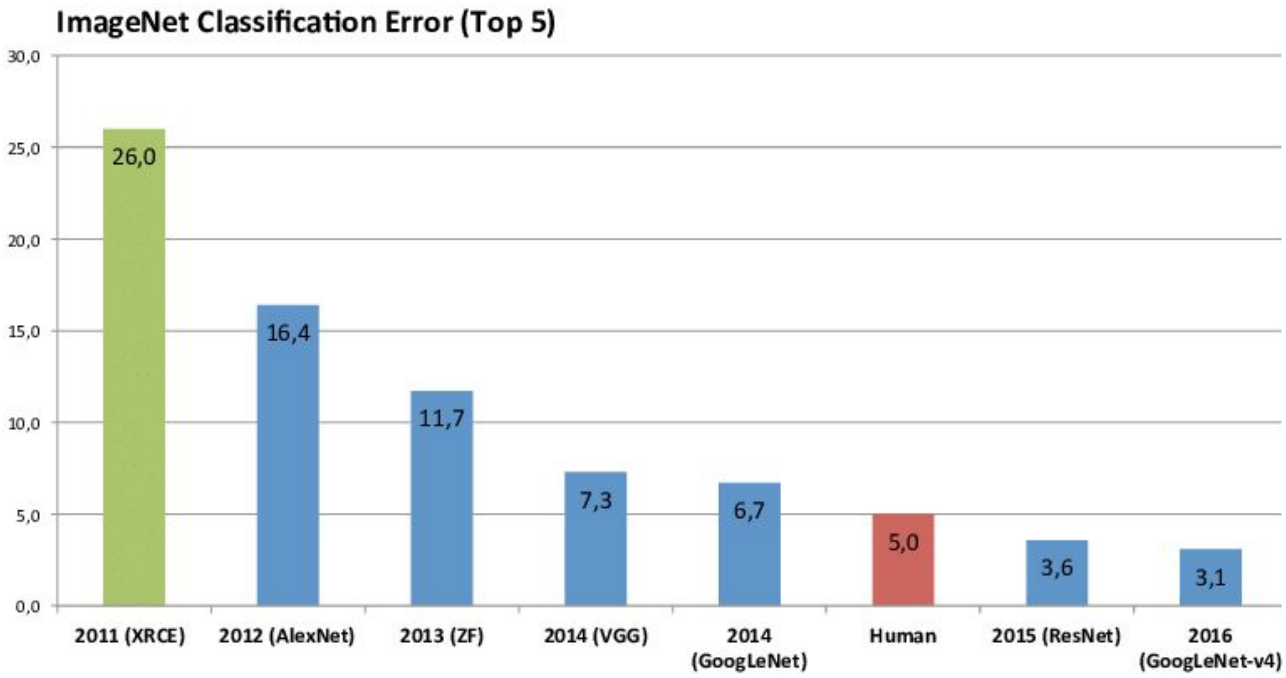
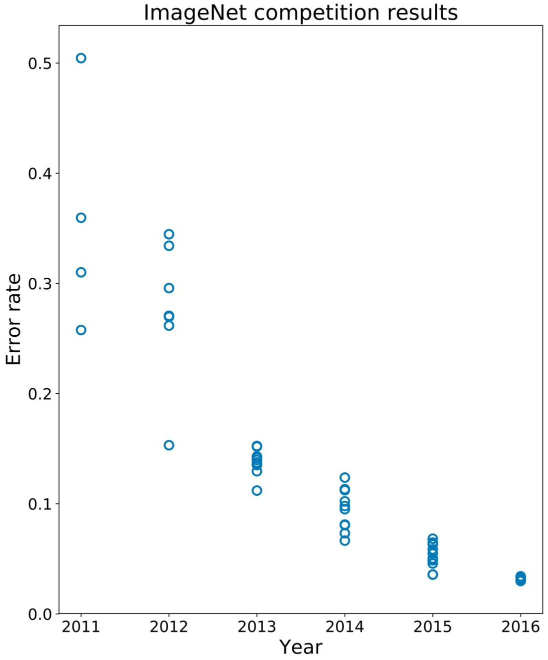


Figure 3: Comparison of the "cat" and "cattle" subtrees between ESP [25] and ImageNet. Within each tree, the size of a node is

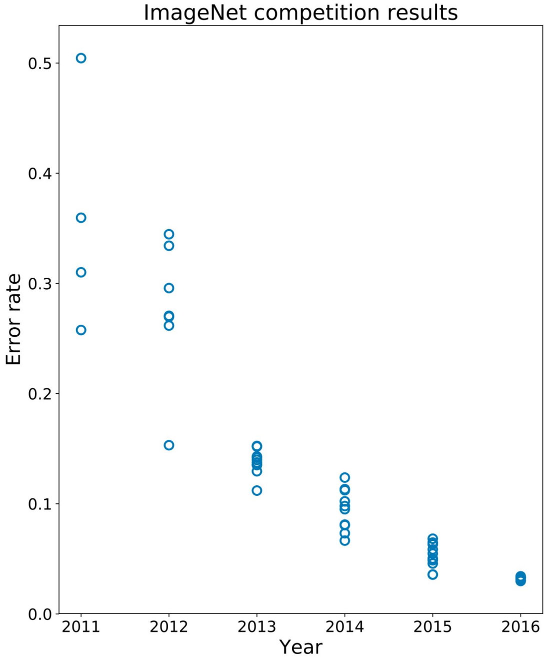
Las CNN funcionan





Las CNN funcionan

- Después de AlexNet en 2012, ningún equipo se equivocó más del 25%
  - Prueba de que las CNN funcionan
  - Mientras se entrene con múltiples GPUs



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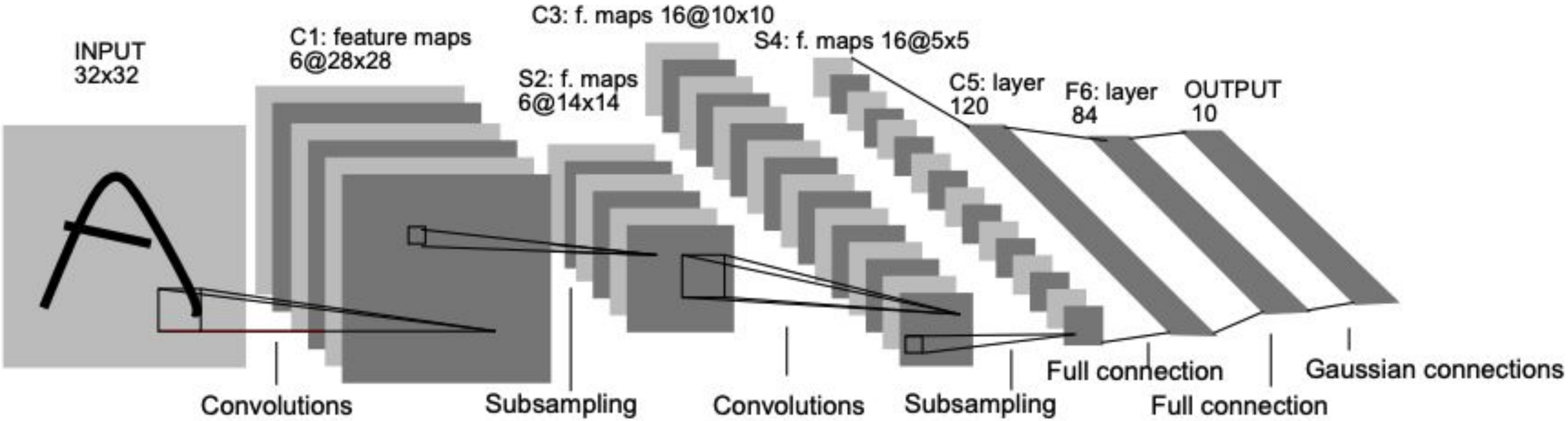
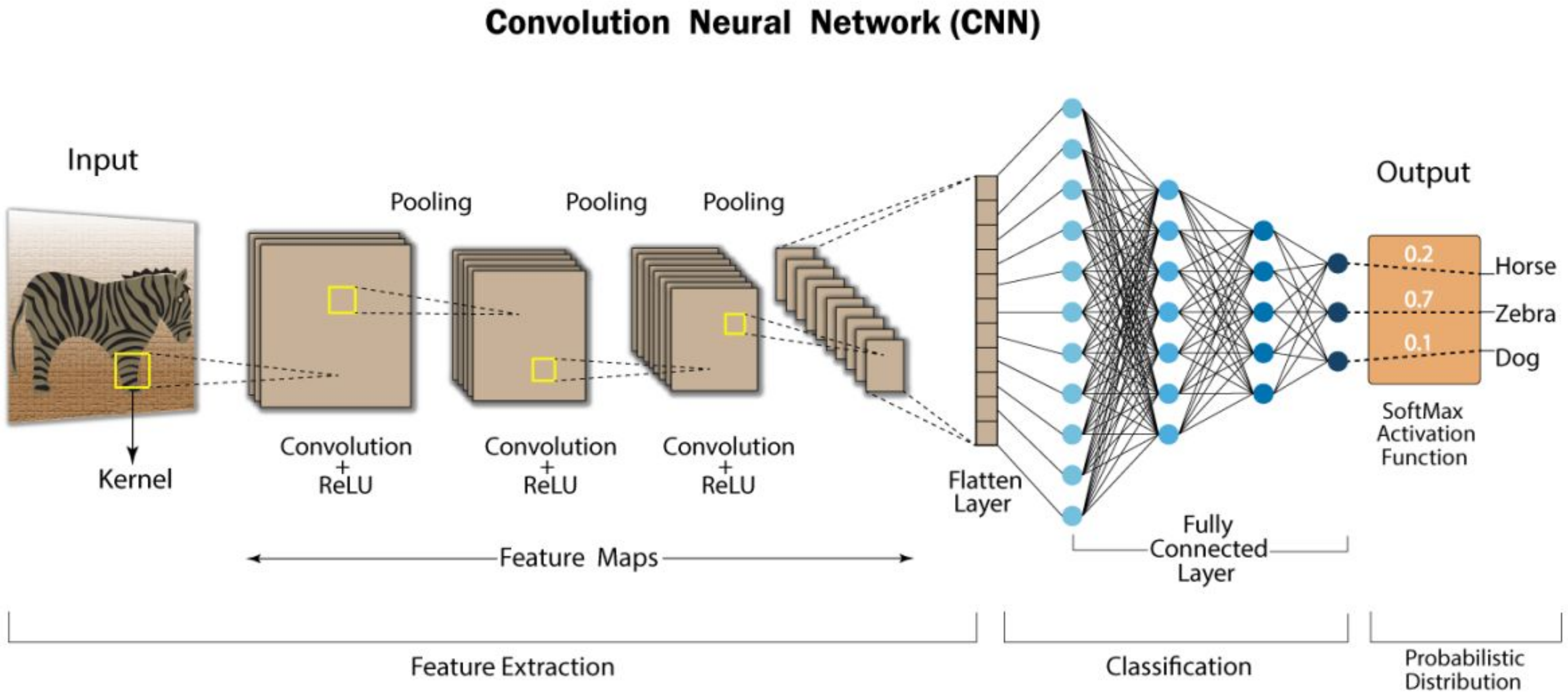
LeNet 5


INPUT
32x32
C1: feature maps
6@28x28
S2: f. maps
6@14x14
C3: f. maps 16@10x10
S4: f. maps 16@5x5
C5: layer
120
F6: layer
84
OUTPUT
10
Convolutions
Subsampling
Convolutions
Subsampling
Full connection
Full connection
Gaussian connections

Arquitectura general



Desafío

- De una imagen grande a color, pasamos a varias imágenes más pequeñas con características extraídas
    - Requiere muchas capas, lo cual complejiza el entrenamiento
    - Redes neuronales de más capas proveen mejores resultados
        - llegaban a límites computacionales hasta 2012

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AlexNet

- Clasifican 1.2 millones de imágenes en 1K clases diferentes.
  - 60M parámetros
  - 650,000 neuronas
  - 5 capas convolucionales
  - 3 capas FC
  - 1K softmax
- Buena implementación de GPU para las conv.
- Non-saturating neurons (ReLu)
- Dropout (0.5)

Preproc: downsampling, normalización de imágenes

Aumentación de imágenes

- Translación de imágenes
- Reflexiones horizontales
- Cambios de intensidades, iluminación
- Cambios de color:
  - PCA en los valores de RGB en training
    - reemplazar por múltiplos de los componentes principales

Hiper parámetros

Mini-batch 128

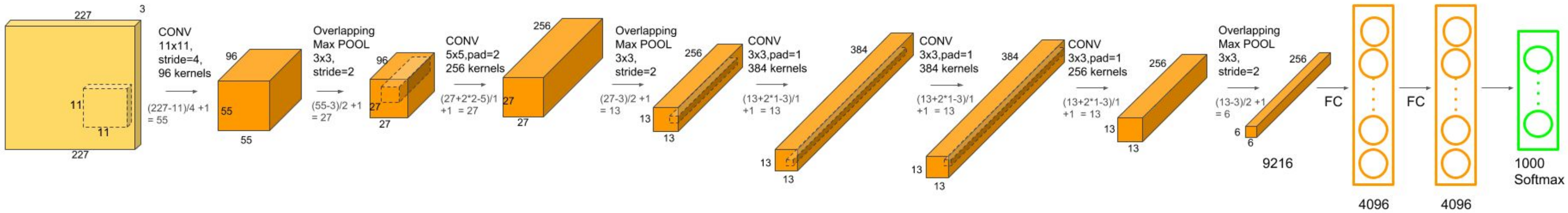
Momentum .9

Decay 0.0005

Inicialización de pesos no nula ~G(0, 0.01)

LR 0.01, dividido por 10 si el validación error no mejora

AlexNet (2012)



AlexNet (2012)

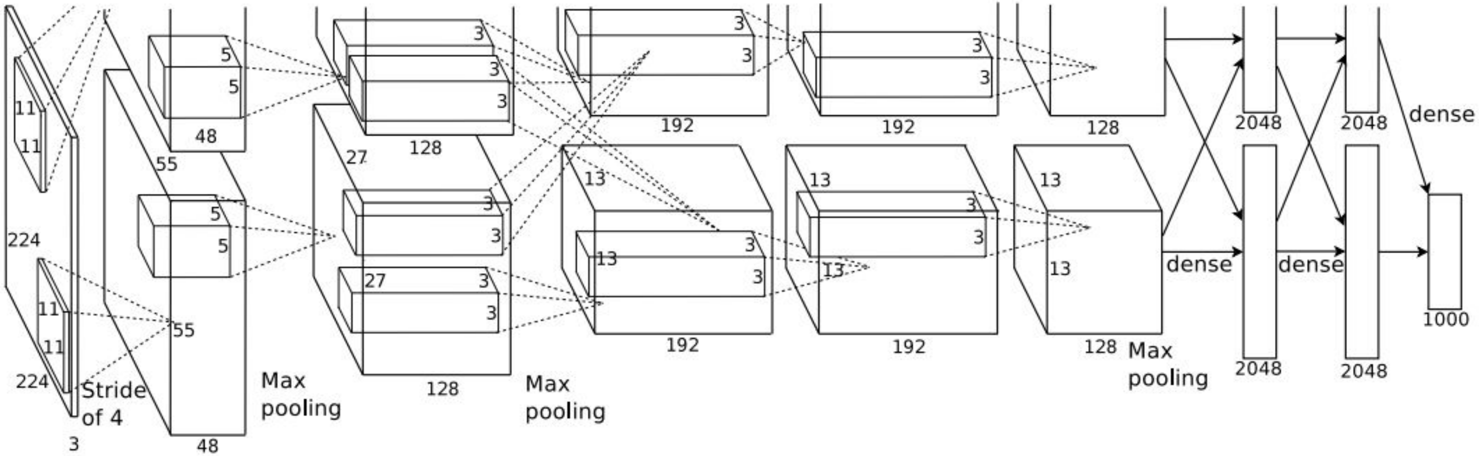
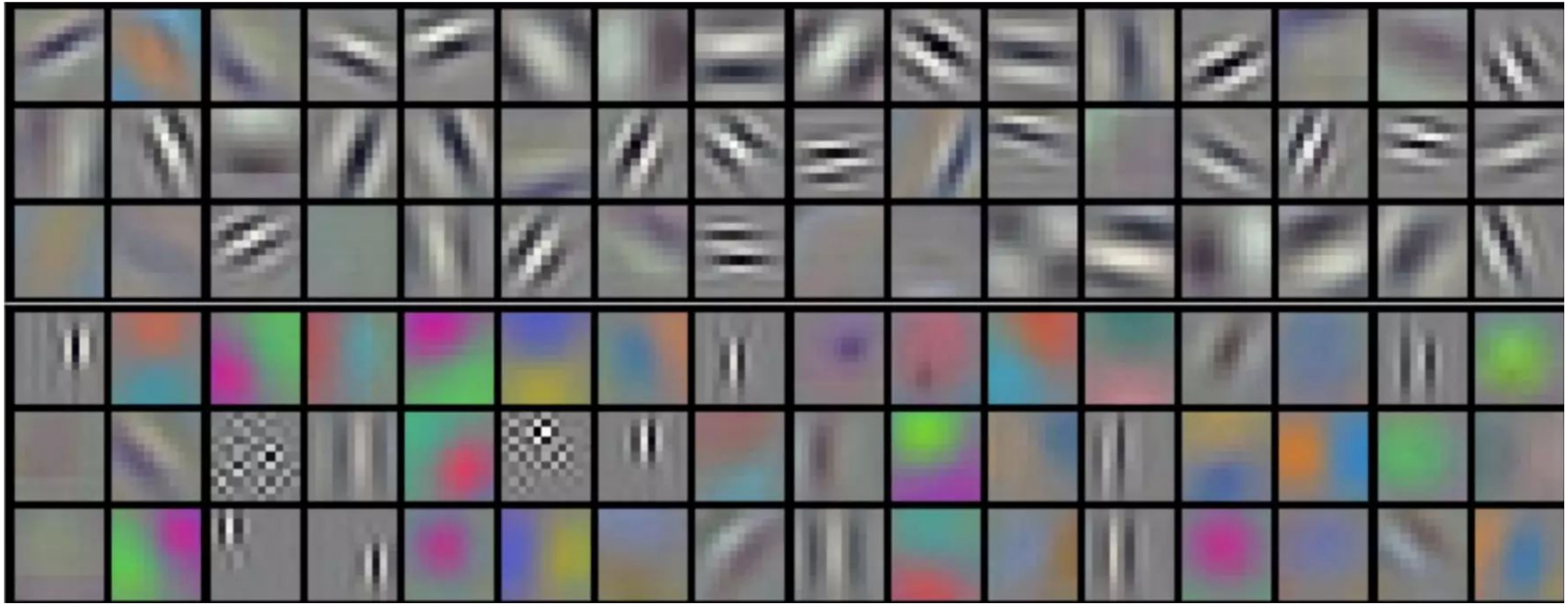


Figure 2: An illustration of the architecture of our CNN, explicitly showing the delineation of responsibilities between the two GPUs. One GPU runs the layer-parts at the top of the figure while the other runs the layer-parts at the bottom. The GPUs communicate only at certain layers. The network's input is 150,528-dimensional, and the number of neurons in the network's remaining layers is given by 253,440–186,624–64,896–64,896–43,264–4096–4096–1000.

Kernels de la primera capa



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VERY DEEP CONVOLUTIONAL NETWORKS FOR LARGE-SCALE IMAGE RECOGNITION

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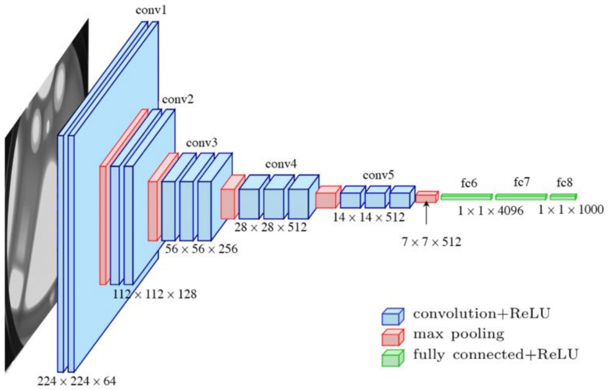
ABSTRACT

In this work we investigate the effect of the convolutional network depth on its accuracy in the large-scale image recognition setting. Our main contribution is a thorough evaluation of networks of increasing depth using an architecture with very small (3×3) convolution filters, which shows that a significant improvement on the prior-art configurations can be achieved by pushing the depth to 16–19 weight layers. These findings were the basis of our ImageNet Challenge 2014 submission, where our team secured the first and the second places in the localisation and classification tracks respectively. We also show that our representations generalise well to other datasets, where they achieve state-of-the-art results. We have made our two best-performing ConvNet models publicly available to facilitate further research on the use of deep visual representations in computer vision.

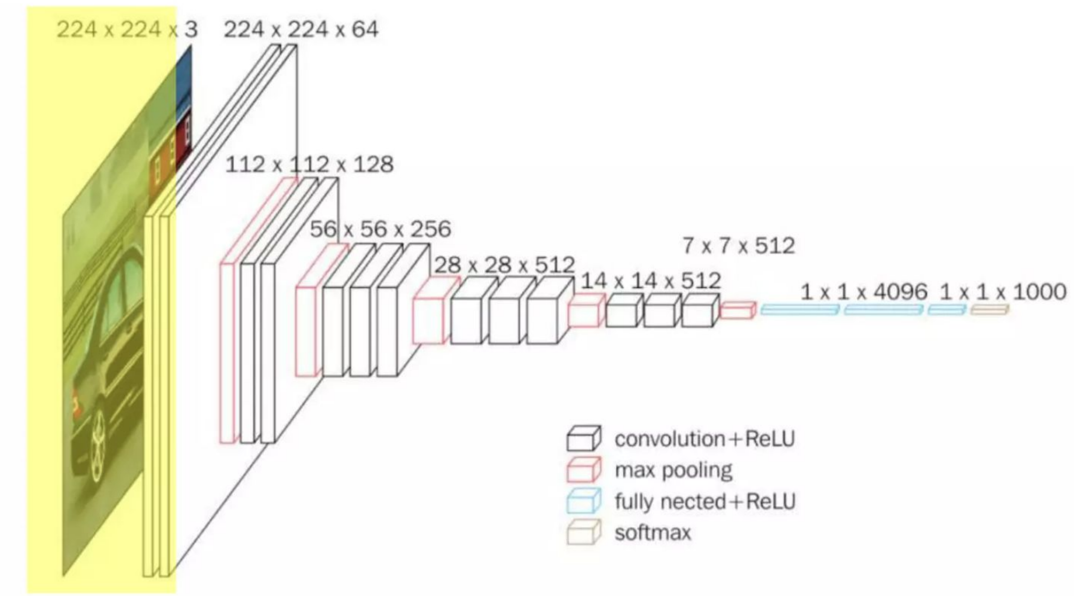
VGGNet

- Querían medir cómo cambiaba la performance a medida que las capas convolucionales crecían.
- Utilizaron ImageNet y varios otros datasets
- Compararon contra otros state-of-the-art algorithms (SOTA).



VGGNet (2014)

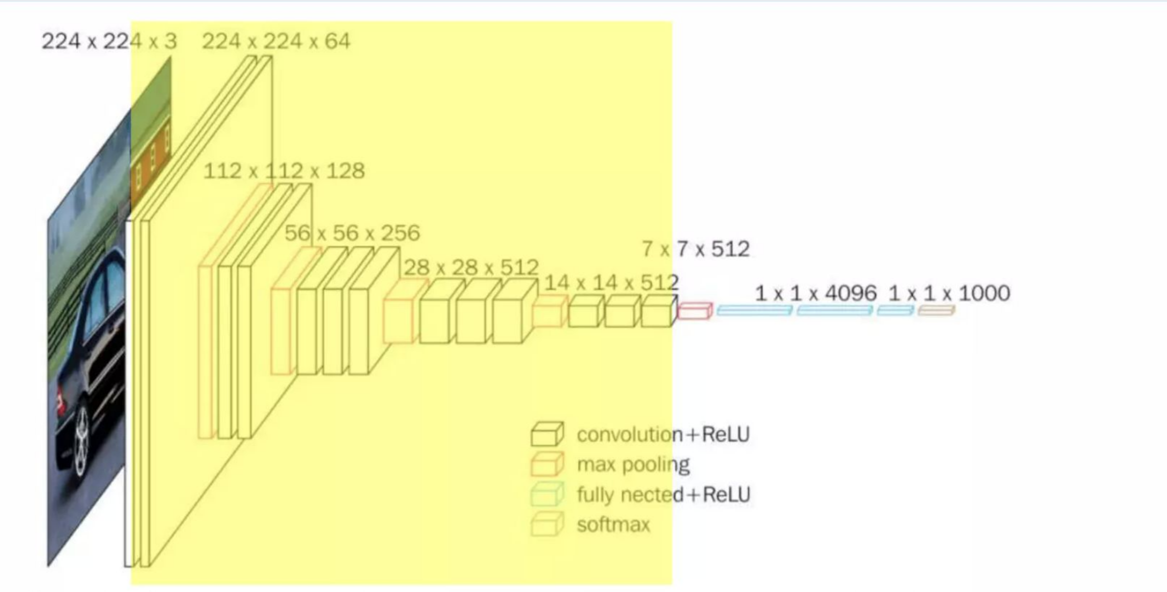
- Preprocessing: subtracting the mean RBG value computed on the training set for each pixel.



VGGNet (2014)

Convolución:

- 3x3 kernel
- 1x1 kernel



Configuraciones

Table 1: **ConvNet configurations** (shown in columns). The depth of the configurations increases from the left (A) to the right (E), as more layers are added (the added layers are shown in bold). The convolutional layer parameters are denoted as "conv⟨receptive field size⟩-⟨number of channels⟩". The ReLU activation function is not shown for brevity.

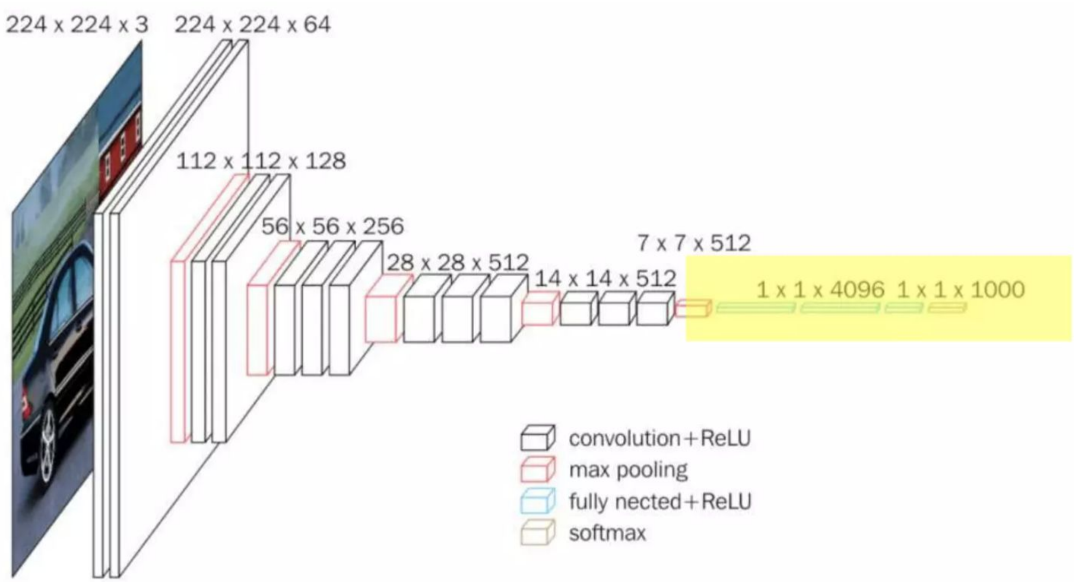
ConvNet Configuration					
A	A-LRN	B	C	D	E
11 weight layers	11 weight layers	13 weight layers	16 weight layers	16 weight layers	19 weight layers
input (224 × 224 RGB image)					
conv3-64	conv3-64 **LRN**	conv3-64 **conv3-64**	conv3-64 conv3-64	conv3-64 conv3-64	conv3-64 conv3-64
maxpool					
conv3-128	conv3-128	conv3-128 **conv3-128**	conv3-128 conv3-128	conv3-128 conv3-128	conv3-128 conv3-128
maxpool					
conv3-256 conv3-256	conv3-256 conv3-256	conv3-256 conv3-256	conv3-256 conv3-256 **conv1-256**	conv3-256 conv3-256 **conv3-256**	conv3-256 conv3-256 conv3-256 **conv3-256**
maxpool					
conv3-512 conv3-512	conv3-512 conv3-512	conv3-512 conv3-512	conv3-512 conv3-512 **conv1-512**	conv3-512 conv3-512 **conv3-512**	conv3-512 conv3-512 conv3-512 **conv3-512**
maxpool					
conv3-512 conv3-512	conv3-512 conv3-512	conv3-512 conv3-512	conv3-512 conv3-512 **conv1-512**	conv3-512 conv3-512 **conv3-512**	conv3-512 conv3-512 conv3-512 **conv3-512**
maxpool					
FC-4096					
FC-4096					
FC-1000					
soft-max					

Table 2: **Number of parameters** (in millions).

Network	A,A-LRN	B	C	D	E
Number of parameters	133	133	134	138	144

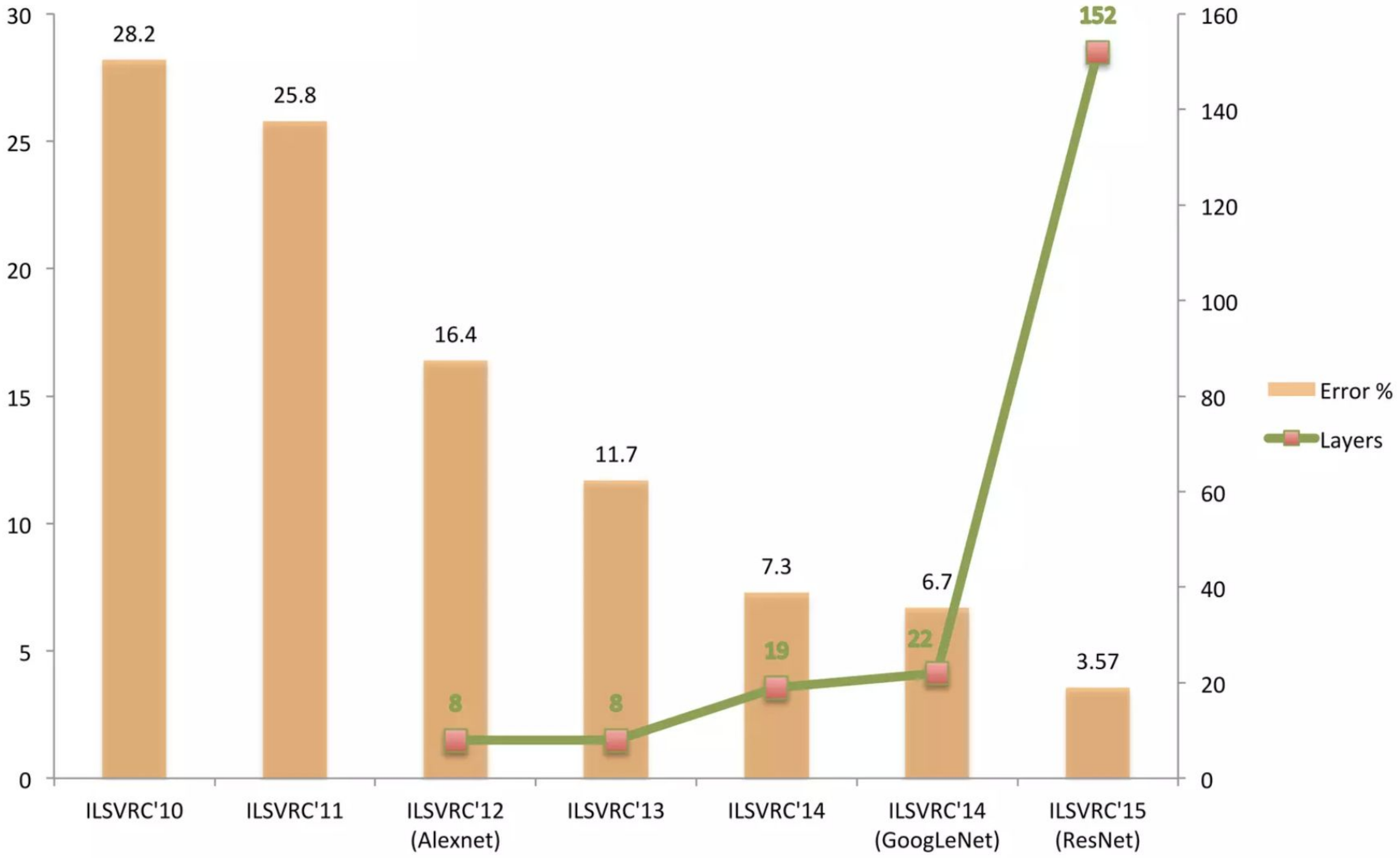
VGGNet (2014)

- 2 FC de 4096 (dropout .5)
- 1 FC de 1000 seguida de una softmax para clasificación multi clases



Training

- Dropout .5 en las 2 primeras FC
- Weight decay (penalidad en la L2 para que los pesos no crezcan mucho)
- A diferencia de AlexNet, no hacen local response normalization
- Los kernels son pequeños para reducir el número de parámetros



30
25
20
15
10
5
0
160
140
120
100
80
60
40
20
0
28.2
25.8
16.4
11.7
7.3
6.7
3.57
8
8
19
22
152
Error %
Layers
ILSVRC'10
ILSVRC'11
ILSVRC'12 (Alexnet)
ILSVRC'13
ILSVRC'14
ILSVRC'14 (GoogLeNet)
ILSVRC'15 (ResNet)

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Going Deeper with Convolutions

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Abstract

We propose a deep convolutional neural network architecture codenamed Inception that achieves the new state of the art for classification and detection in the ImageNet Large-Scale Visual Recognition Challenge 2014 (ILSVRC14). The main hallmark of this architecture is the improved utilization of the computing resources inside the network. By a carefully crafted design, we increased the depth and width of the network while keeping the computational budget constant. To optimize quality, the architectural decisions were based on the Hebbian principle and the intuition of multi-scale processing. One particular incarnation used in our submission for ILSVRC14 is called GoogLeNet, a 22 layers deep network, the quality of which is assessed in the context of classification and detection.

ger and bigger deep networks, but from the synergy of deep architectures and classical computer vision, like the R-CNN algorithm by Girshick et al [6].

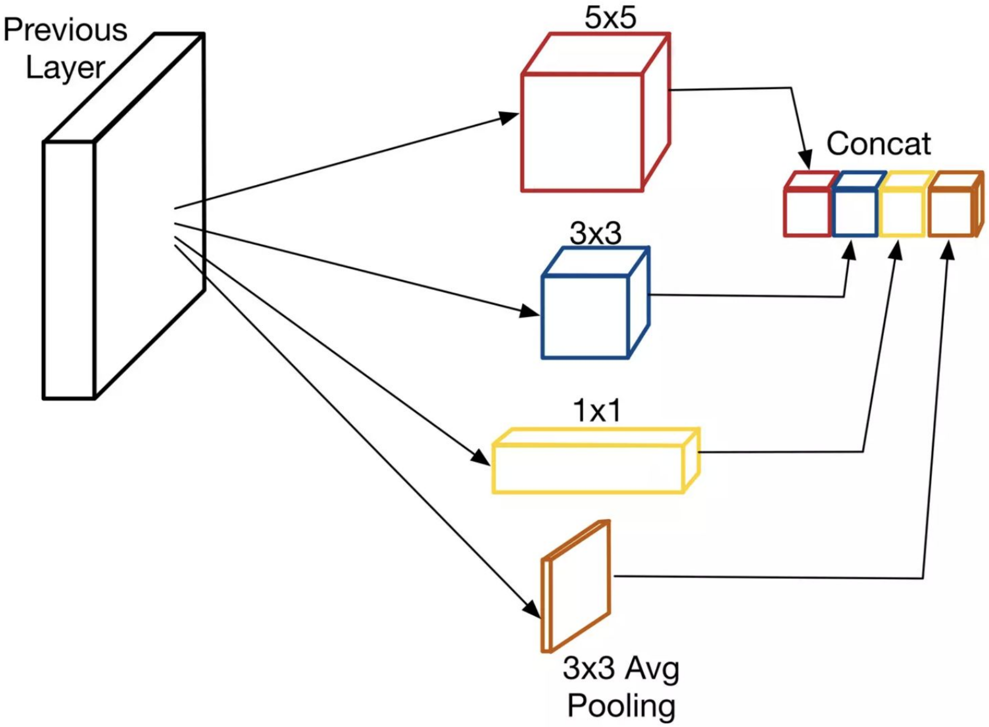
Another notable factor is that with the ongoing traction of mobile and embedded computing, the efficiency of our algorithms – especially their power and memory use – gains importance. It is noteworthy that the considerations leading to the design of the deep architecture presented in this paper included this factor rather than having a sheer fixation on accuracy numbers. For most of the experiments, the models were designed to keep a computational budget of 1.5 billion multiply-adds at inference time, so that the they do not end up to be a purely academic curiosity, but could be put to real world use, even on large datasets, at a reasonable cost.

In this paper, we will focus on an efficient deep neural network architecture for computer vision, codenamed Inception, which derives its name from the Network in net-

GoogLeNet

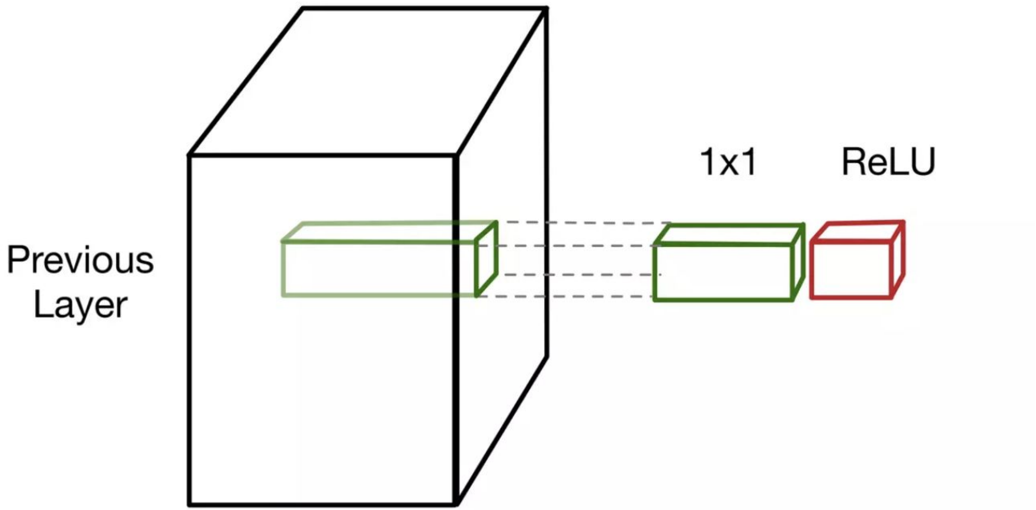
- vs. AlexNet
  - + rápido
  - + eficiente
  - - memoria
- Aplica varios filtros al mismo patch y concatena los outputs en un vector de salida:
  - 1x1
  - 3x3
  - 5x5

Inception - naive

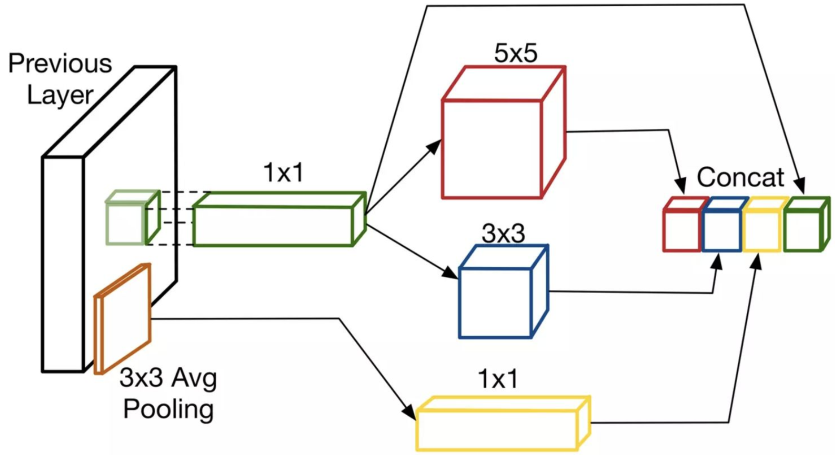


1x1 capa de reducción

- Antes de realizar las convoluciones 3x3 y 5x5 para eficiencia

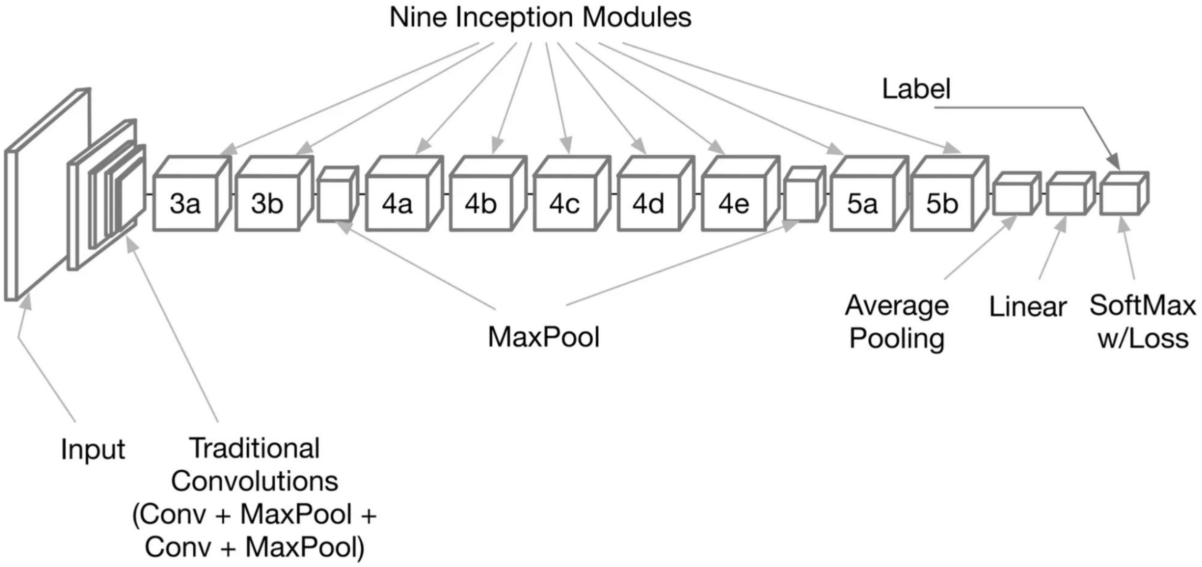


1x1 capa de reducción



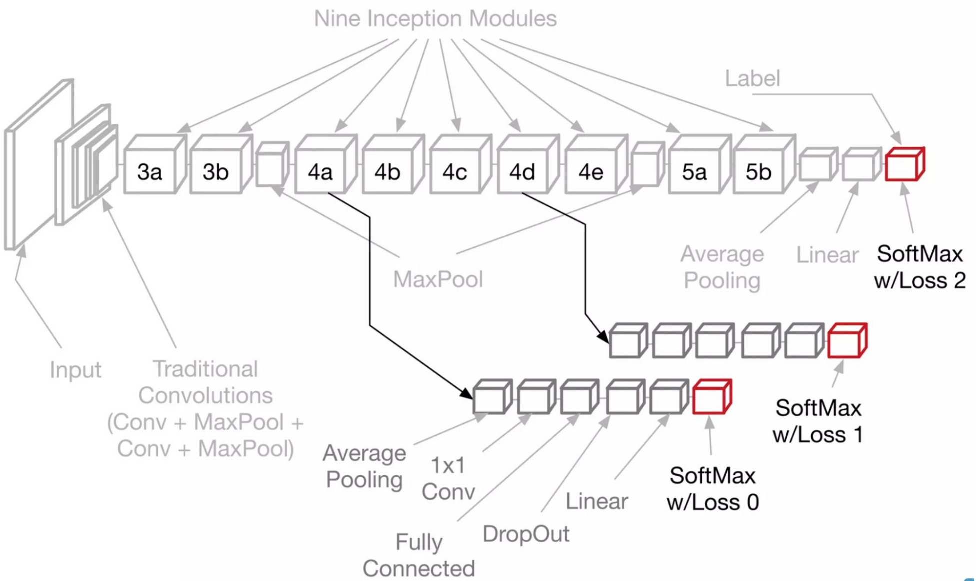
Apilado de capas de inception

- Max pool con stride 2 para disminuir el tamaño por dos
  - Es una forma de apilar capas de inception (conv) sin incrementar tanto la cantidad de parámetros.



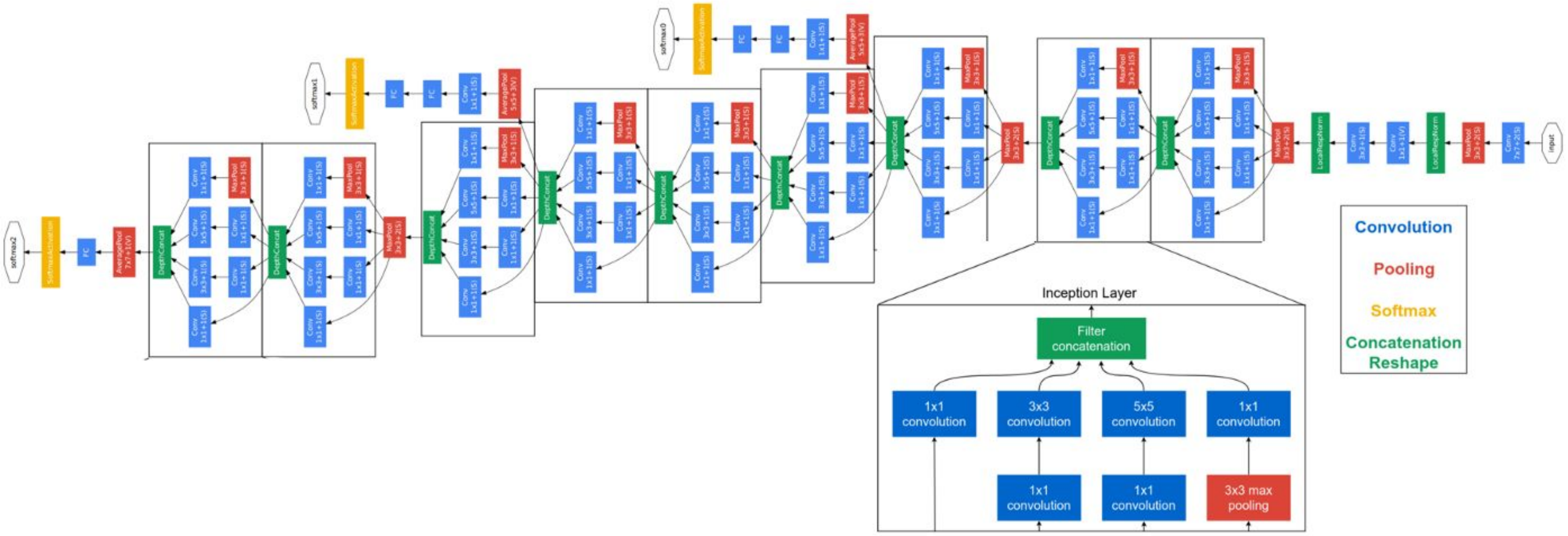
Vanishing gradient

- Para evitar el vanishing gradient en la profundidad se agregan loss.
  - En entrenamiento las loss se suman (factor .3 para la intermediarias)
- Average Pooling en vez de FC
  - Evita overfitting



GoogLeNet (2014)

Introduced multiple branches with different filter sizes in a single layer known as "inception layer"



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Deep Residual Learning for Image Recognition

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Abstract

Deeper neural networks are more difficult to train. We present a residual learning framework to ease the training of networks that are substantially deeper than those used previously. We explicitly reformulate the layers as learning residual functions with reference to the layer inputs, instead of learning unreferenced functions. We provide comprehensive empirical evidence showing that these residual networks are easier to optimize, and can gain accuracy from considerably increased depth. On the ImageNet dataset we evaluate residual nets with a depth of up to 152 layers—8× deeper than VGG nets [41] but still having lower complexity. An ensemble of these residual nets achieves 3.57% error on the ImageNet test *set. This result won the 1st place on the ILSVRC 2015 classification task. We also present analysis on CIFAR-10 with 100 and 1000 layers.*

The depth of representations is of central importance for many visual recognition tasks. Solely due to our extremely deep representations, we obtain a 28% relative improvement on the COCO object detection dataset. Deep residual nets are foundations of our submissions to ILSVRC & COCO 2015 competitions[1]*, where we also won the 1st places on the tasks of ImageNet detection, ImageNet localization, COCO detection, and COCO segmentation.*

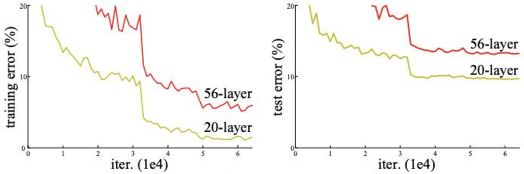


Figure 1. Training error (left) and test error (right) on CIFAR-10 with 20-layer and 56-layer "plain" networks. The deeper network has higher training error, and thus test error. Similar phenomena on ImageNet is presented in Fig. 4.

greatly benefited from very deep models.

Driven by the significance of depth, a question arises: *Is learning better networks as easy as stacking more layers?* An obstacle to answering this question was the notorious problem of vanishing/exploding gradients [1, 9], which hamper convergence from the beginning. This problem, however, has been largely addressed by normalized initialization [23, 9, 37, 13] and intermediate normalization layers [16], which enable networks with tens of layers to start converging for stochastic gradient descent (SGD) with backpropagation [22].

When deeper networks are able to start converging, a *degradation* problem has been exposed: with the network

Más capas mejor rendim…

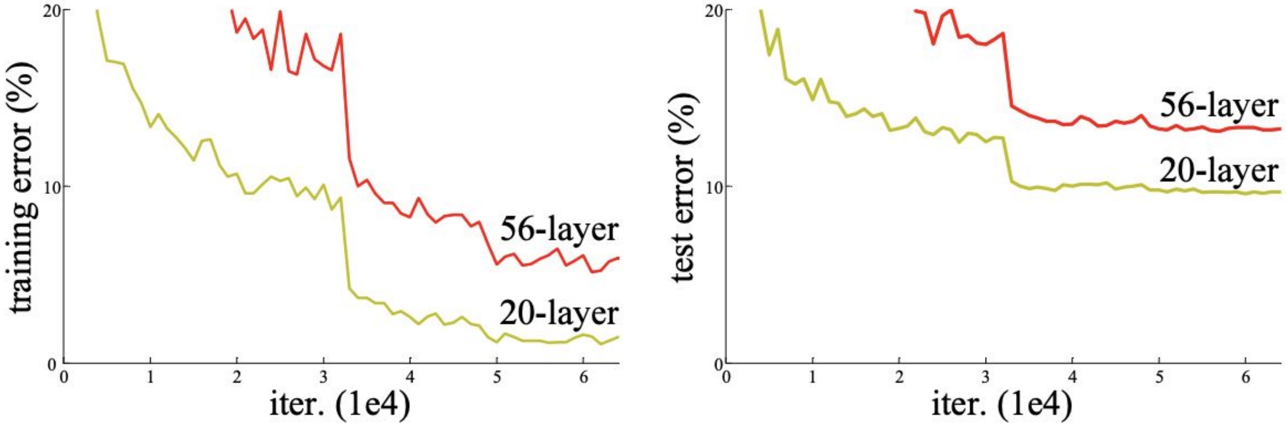


Figure 1. Training error (left) and test error (right) on CIFAR-10 with 20-layer and 56-layer "plain" networks. The deeper network has higher training error, and thus test error. Similar phenomena on ImageNet is presented in Fig. 4.

Idea

- Redes convolucionales profundas
  - Incrementa el error en la profundidad de capas
- Saltear conexiones
  - vanishing gradient
  - disminuir tiempo computacional
  - mejorar el rendimiento

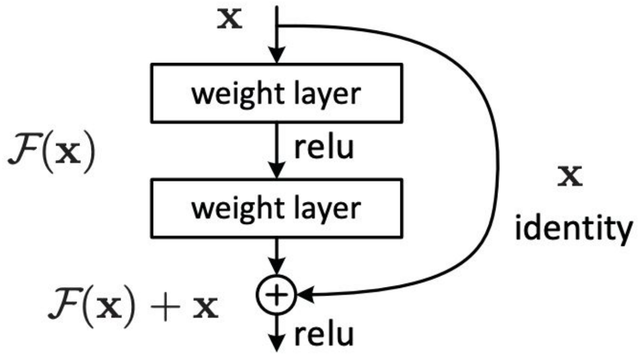
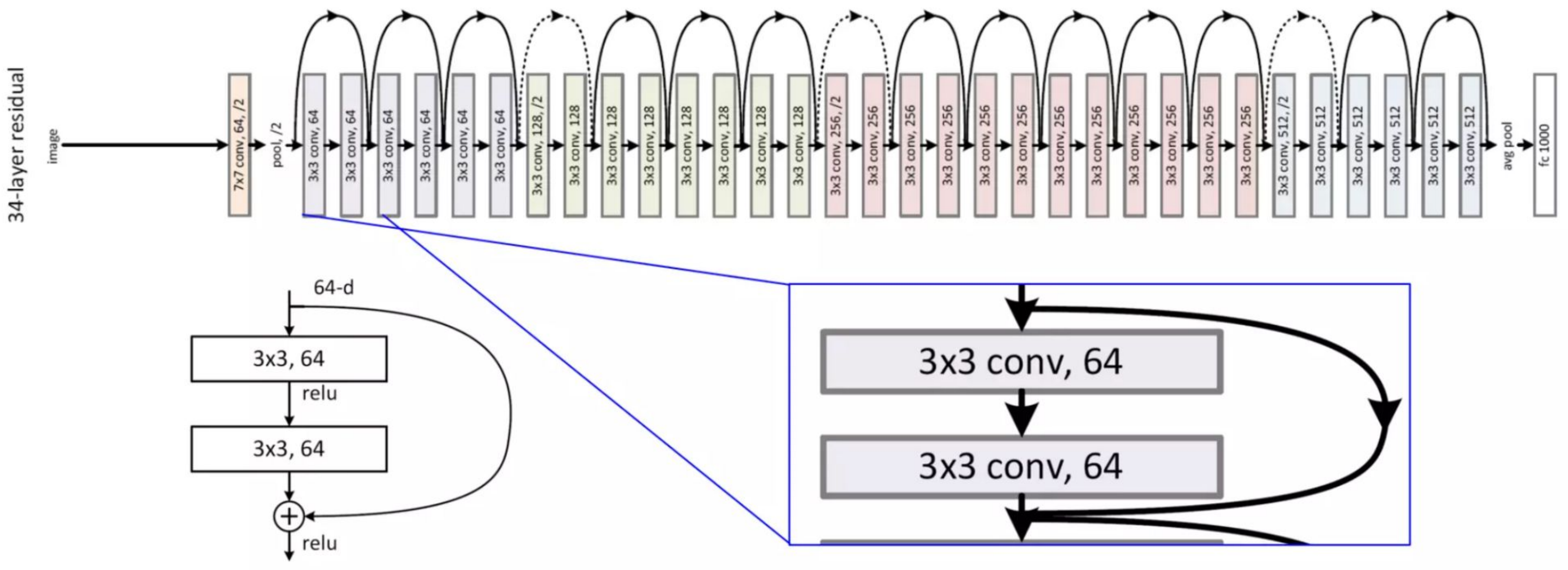
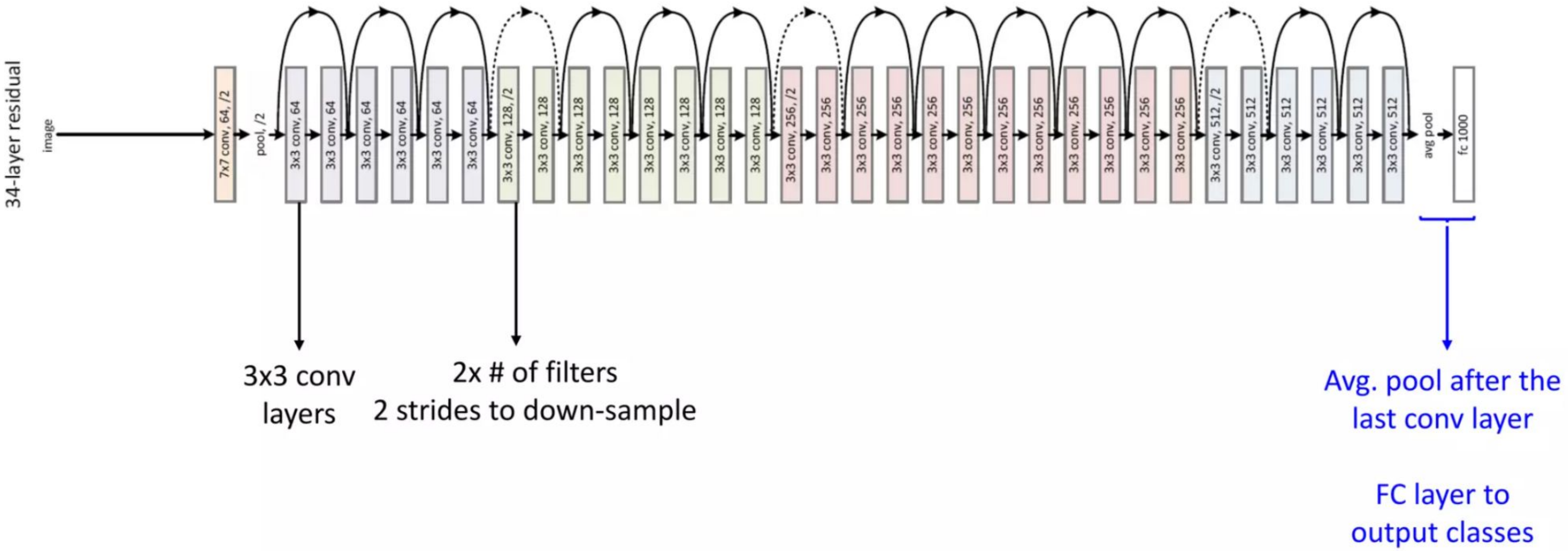


Figure 2. Residual learning: a building block.

Bloques residuales apilados



Bloques residuales apilados



Arquitecturas evaluadas

layer name	output size	18-layer	34-layer	50-layer	101-layer	152-layer
conv1	112×112	7×7, 64, stride 2				
conv2_x	56×56	3×3 max pool, stride 2				
		$\left[\begin{array}{c}3\times3, 64\\ 3\times3, 64\end{array}\right]\times2$	$\left[\begin{array}{c}3\times3, 64\\ 3\times3, 64\end{array}\right]\times3$	$\left[\begin{array}{c}1\times1, 64\\ 3\times3, 64\\ 1\times1, 256\end{array}\right]\times3$	$\left[\begin{array}{c}1\times1, 64\\ 3\times3, 64\\ 1\times1, 256\end{array}\right]\times3$	$\left[\begin{array}{c}1\times1, 64\\ 3\times3, 64\\ 1\times1, 256\end{array}\right]\times3$
conv3_x	28×28	$\left[\begin{array}{c}3\times3, 128\\ 3\times3, 128\end{array}\right]\times2$	$\left[\begin{array}{c}3\times3, 128\\ 3\times3, 128\end{array}\right]\times4$	$\left[\begin{array}{c}1\times1, 128\\ 3\times3, 128\\ 1\times1, 512\end{array}\right]\times4$	$\left[\begin{array}{c}1\times1, 128\\ 3\times3, 128\\ 1\times1, 512\end{array}\right]\times4$	$\left[\begin{array}{c}1\times1, 128\\ 3\times3, 128\\ 1\times1, 512\end{array}\right]\times8$
conv4_x	14×14	$\left[\begin{array}{c}3\times3, 256\\ 3\times3, 256\end{array}\right]\times2$	$\left[\begin{array}{c}3\times3, 256\\ 3\times3, 256\end{array}\right]\times6$	$\left[\begin{array}{c}1\times1, 256\\ 3\times3, 256\\ 1\times1, 1024\end{array}\right]\times6$	$\left[\begin{array}{c}1\times1, 256\\ 3\times3, 256\\ 1\times1, 1024\end{array}\right]\times23$	$\left[\begin{array}{c}1\times1, 256\\ 3\times3, 256\\ 1\times1, 1024\end{array}\right]\times36$
conv5_x	7×7	$\left[\begin{array}{c}3\times3, 512\\ 3\times3, 512\end{array}\right]\times2$	$\left[\begin{array}{c}3\times3, 512\\ 3\times3, 512\end{array}\right]\times3$	$\left[\begin{array}{c}1\times1, 512\\ 3\times3, 512\\ 1\times1, 2048\end{array}\right]\times3$	$\left[\begin{array}{c}1\times1, 512\\ 3\times3, 512\\ 1\times1, 2048\end{array}\right]\times3$	$\left[\begin{array}{c}1\times1, 512\\ 3\times3, 512\\ 1\times1, 2048\end{array}\right]\times3$
	1×1	average pool, 1000-d fc, softmax				
FLOPs		1.8×10^9	3.6×10^9	3.8×10^9	7.6×10^9	11.3×10^9

Table 1. Architectures for ImageNet. Building blocks are shown in brackets (see also Fig. 5), with the numbers of blocks stacked. Down-sampling is performed by conv3_1, conv4_1, and conv5_1 with a stride of 2.

Error test, plain net vs ResNet

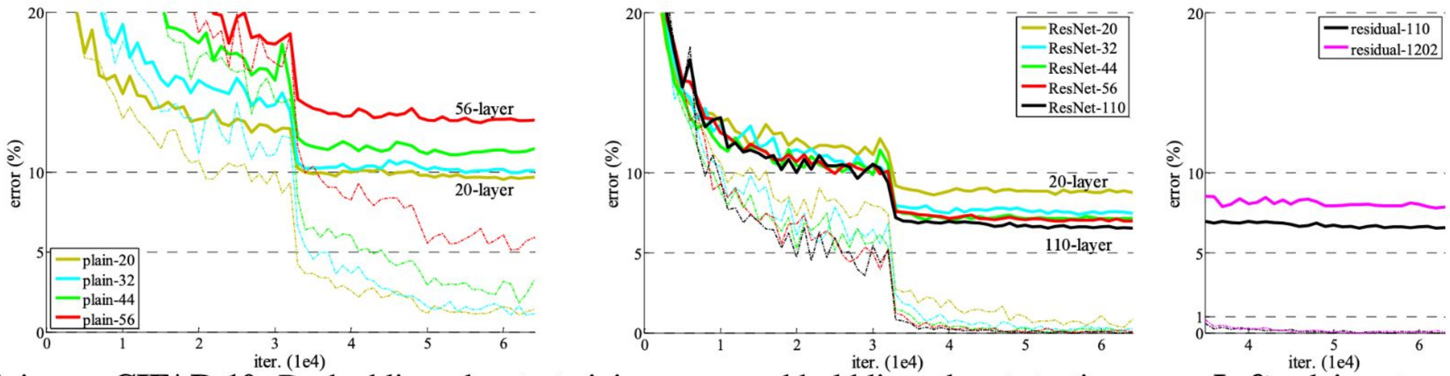
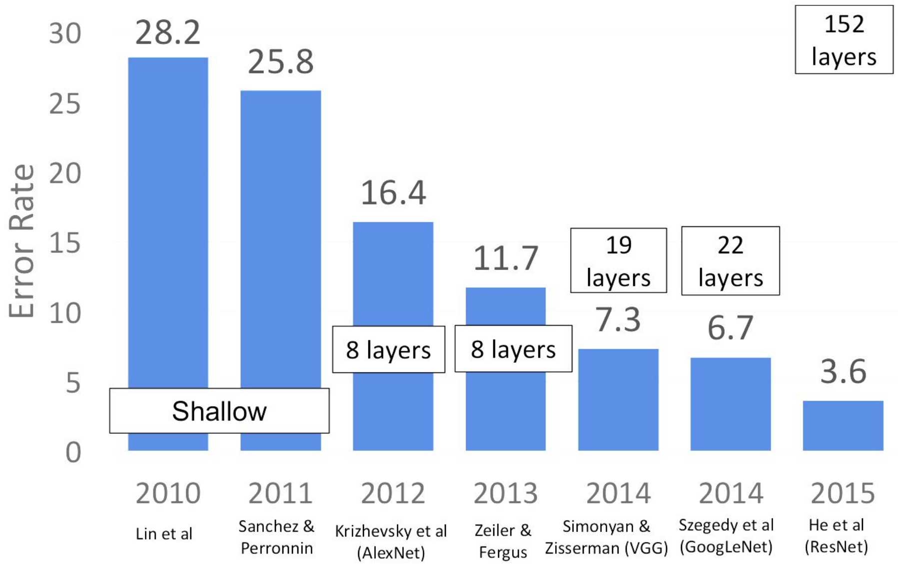


Figure 6. Training on **CIFAR-10**. Dashed lines denote training error, and bold lines denote testing error. **Left**: plain networks. The error of plain-110 is higher than 60% and not displayed. **Middle**: ResNets. **Right**: ResNets with 110 and 1202 layers.

Go Deep or Go Home

- Redes profundas puede aprender mejores representaciones de la entrada.

ImageNet classification challenge



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1. Clasificación de imágenes e ImageNet challenge
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Visualizing and Understanding Convolutional Networks

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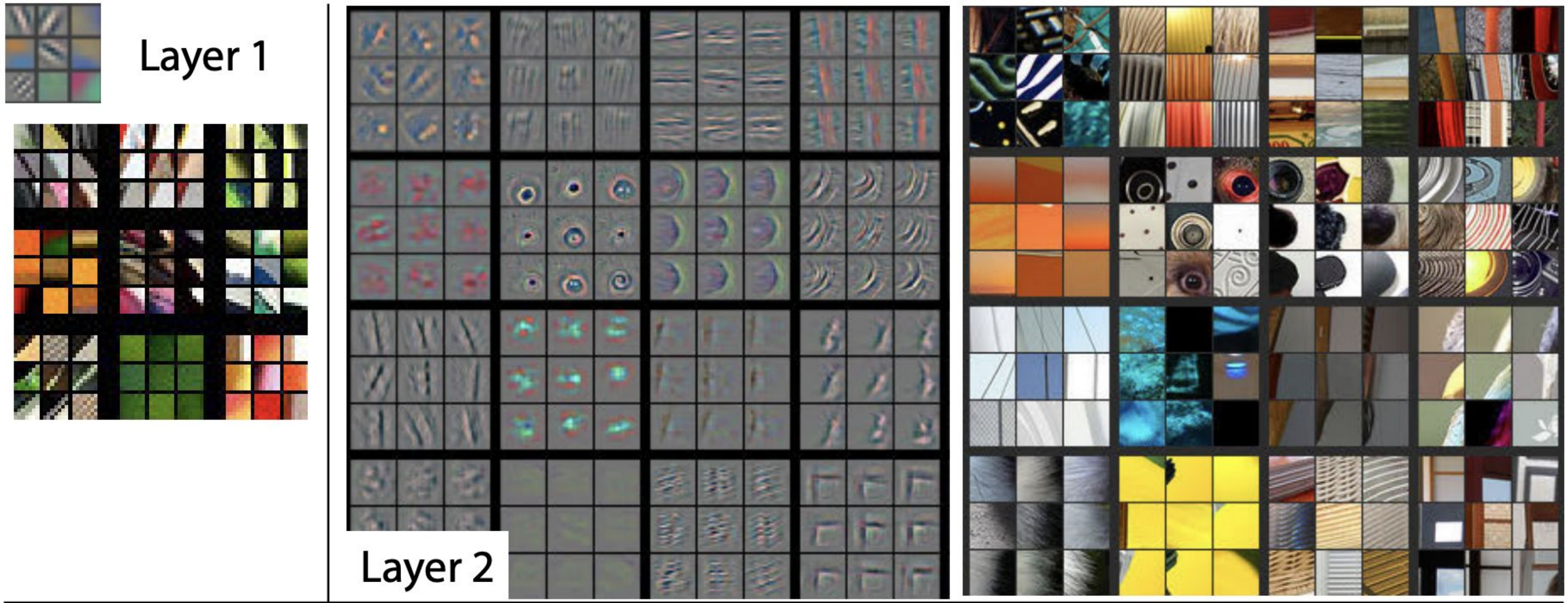
Abstract

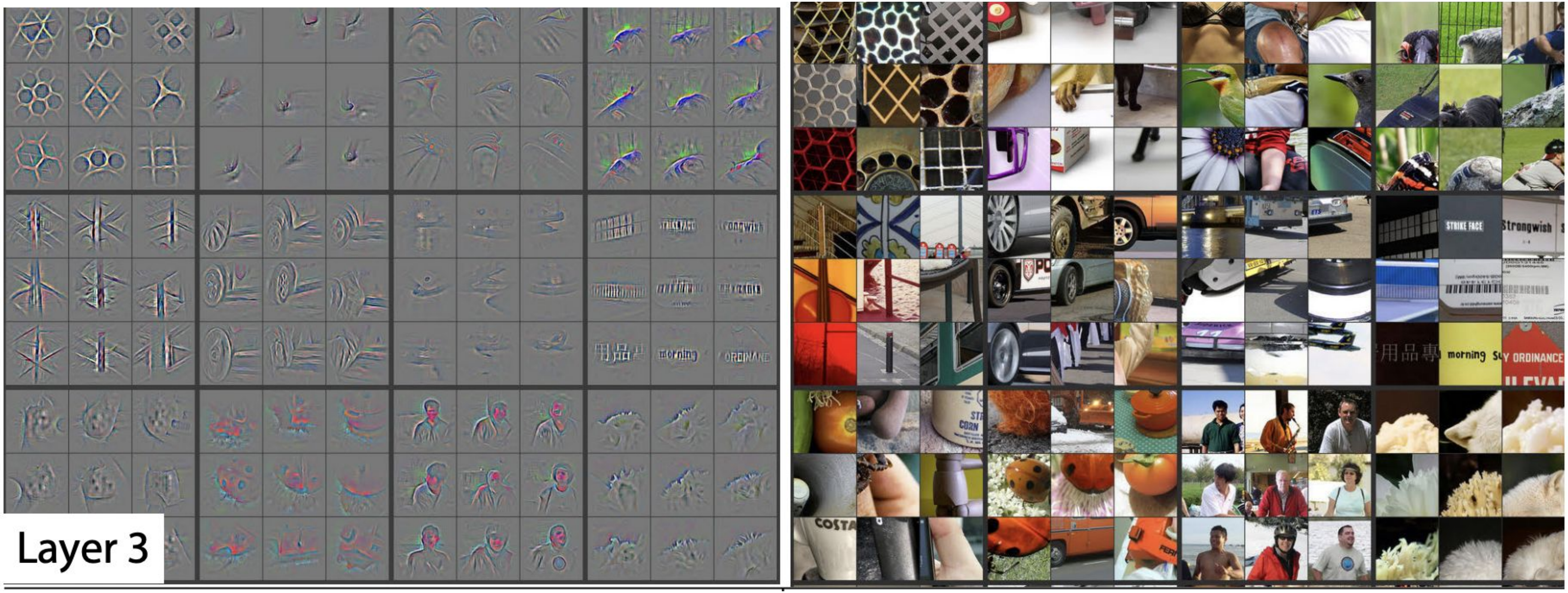
Large Convolutional Network models have recently demonstrated impressive classification performance on the ImageNet benchmark (Krizhevsky et al., 2012). However there is no clear understanding of why they perform so well, or how they might be improved. In this paper we address both issues. We introduce a novel visualization technique that gives insight into the function of intermediate feature layers and the operation of the classifier. Used in a diagnostic role, these visualizations allow us to find model architectures that outperform Krizhevsky *et al.* on the ImageNet classification benchmark. We also perform an ablation study to discover the performance contribution from different model layers. We show our ImageNet model generalizes well to other datasets: when the softmax classifier is retrained, it convincingly beats the current state-of-the-art results on Caltech-101 and Caltech-256 datasets.

est in convnet models: (i) the availability of much larger training sets, with millions of labeled examples; (ii) powerful GPU implementations, making the training of very large models practical and (iii) better model regularization strategies, such as Dropout (Hinton et al., 2012).

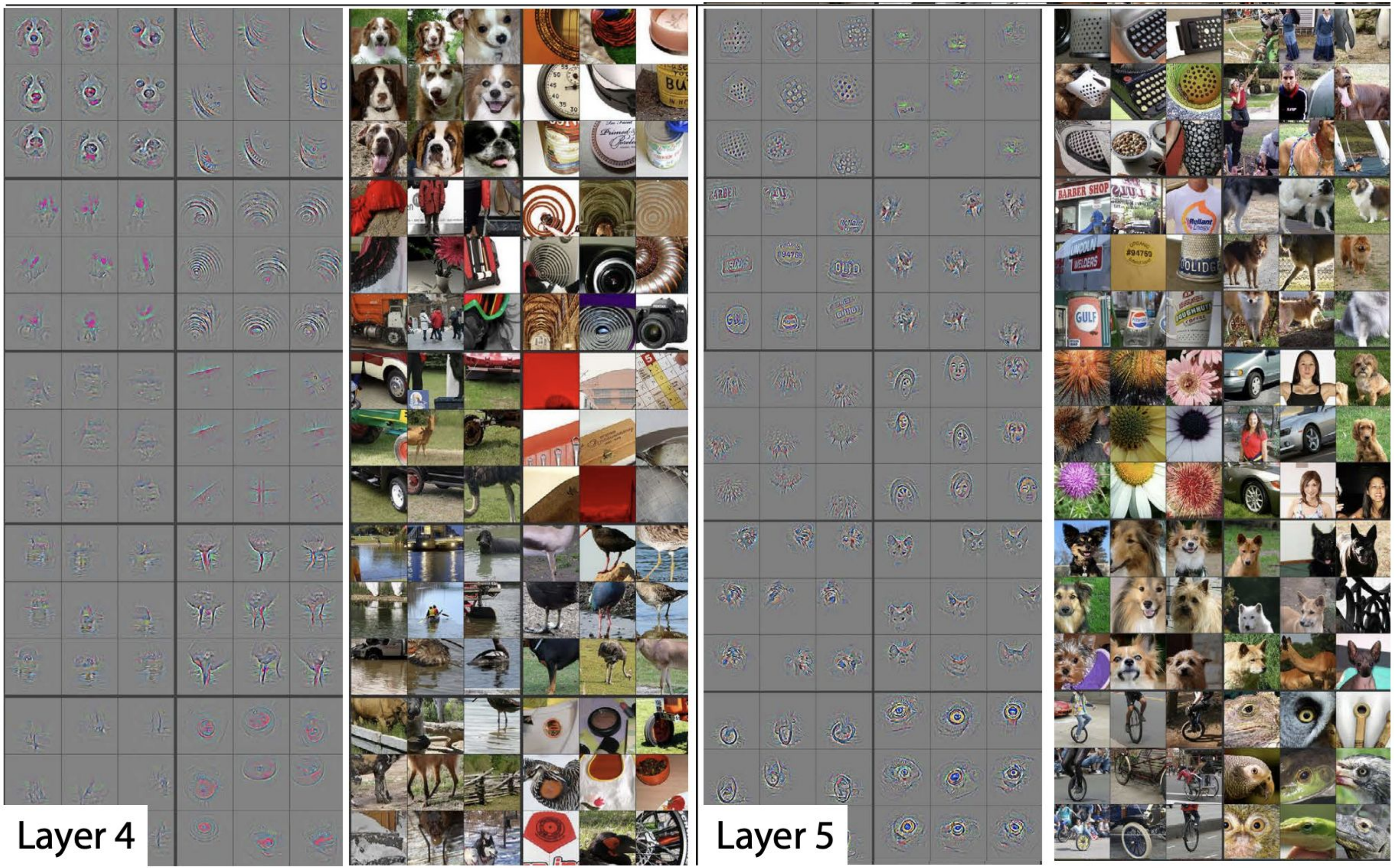
Despite this encouraging progress, there is still little insight into the internal operation and behavior of these complex models, or how they achieve such good performance. From a scientific standpoint, this is deeply unsatisfactory. Without clear understanding of how and why they work, the development of better models is reduced to trial-and-error. In this paper we introduce a visualization technique that reveals the input stimuli that excite individual feature maps at any layer in the model. It also allows us to observe the evolution of features during training and to diagnose potential problems with the model. The visualization technique we propose uses a multi-layered Deconvolutional Network (deconvnet), as proposed by (Zeiler et al., 2011), to project the feature activations back to the input pixel space. We also perform a sensitivity

Figure 2. Visualization of features in a fully trained model. For layers 2-5 we show the top 9 activations in a random subset of feature maps across the validation data, projected down to pixel space using our deconvolutional network approach. Our reconstructions are *not* samples from the model: they are reconstructed patterns from the validation set that cause high activations in a given feature map. For each feature map we also show the corresponding image patches. Note: (i) the the strong grouping within each feature map, (ii) greater invariance at higher layers and (iii) exaggeration of discriminative parts of the image, e.g. eyes and noses of dogs (layer 4, row 1, cols 1). Best viewed in electronic form.





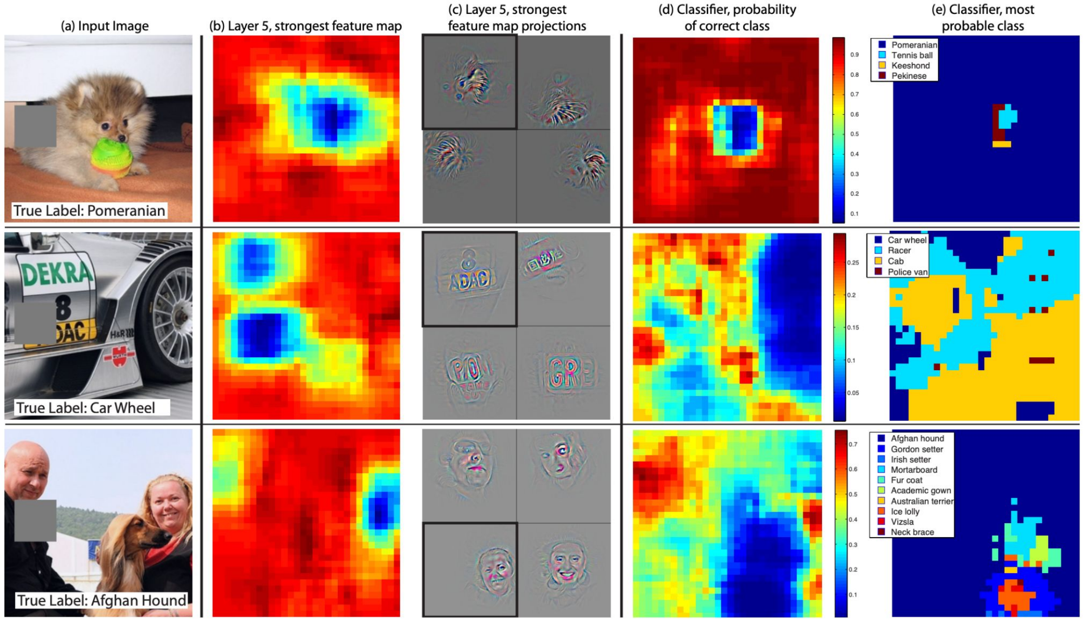
Layer 3


Layer 4
Layer 5
BARBER SHOP
GULF

Occlusion sensitivity/ sensibilidad a la oclusión

- Cómo afecta a la predicción tapar un pedacito de la imagen.
    - Cuando tapo la cara del perro, dejo de clasificar al perro como Pomeranian.

→ la cara es muy importante!



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Is object localization for free? – Weakly-supervised learning with convolutional neural networks

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Léon Bottou† MSR, New York, USA
Ivan Laptev* INRIA, Paris, France
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Abstract

Successful methods for visual object recognition typically rely on training datasets containing lots of richly annotated images. Detailed image annotation, e.g. by object bounding boxes, however, is both expensive and often subjective. We describe a weakly supervised convolutional neural network (CNN) for object classification that relies only on image-level labels, yet can learn from cluttered scenes containing multiple objects. We quantify its object classification and object location prediction performance on the Pascal VOC 2012 (20 object classes) and the much larger Microsoft COCO (80 object classes) datasets. We find that the network (i) outputs accurate image-level labels, (ii) predicts approximate locations (but not extents) of objects, and (iii) performs comparably to its fully-supervised counterparts using object bounding box annotation for training.

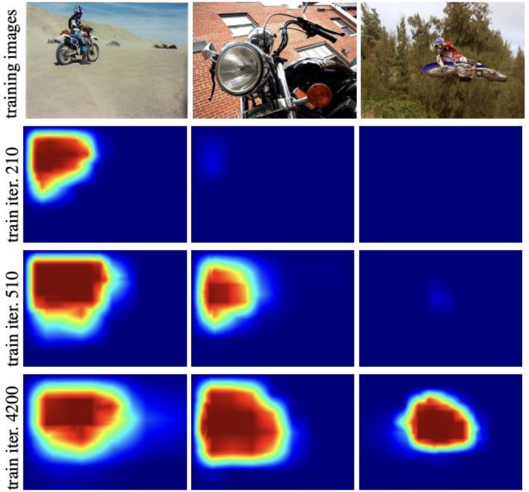


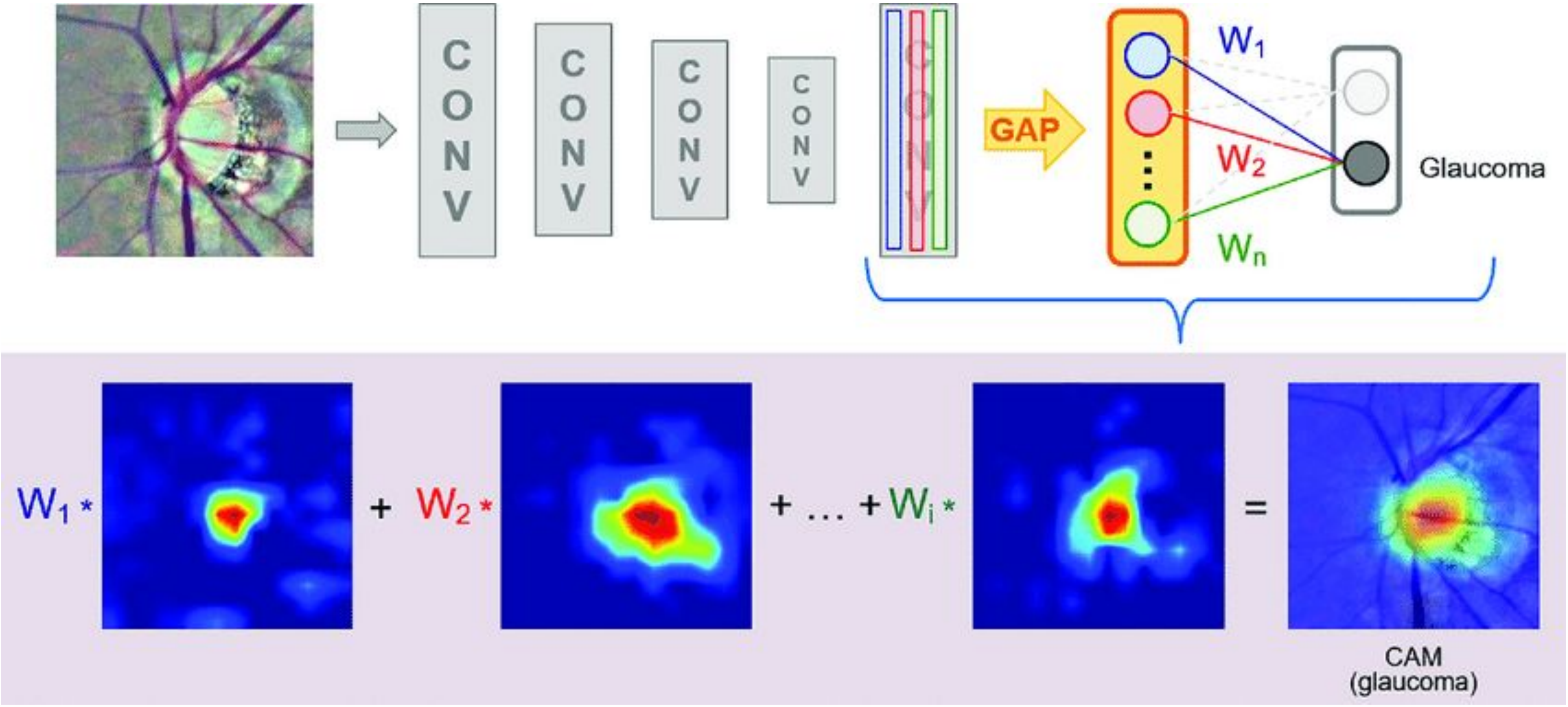
Figure 1: Evolution of localization score maps for the `motorbike` class over iterations of our weakly-supervised

1. Introduction

Visual object recognition entails much more than deter-

Cam

- Se utiliza para destacar las áreas relevantes de una imagen en función de la clase que se está buscando.
- Durante el entrenamiento, la CNN aprende a asignar pesos a diferentes feature maps y regiones de la imagen.
  - Estos pesos se utilizan para ponderar las activaciones de las capas convolucionales finales de la red.
  - La suma ponderada de las activaciones se utiliza para generar un mapa de activación de clase que resalta las regiones importantes para la clasificación de una clase en particular.



Global Average Pooling (GAP)

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Grad-CAM:
Visual Explanations from Deep Networks via Gradient-based Localization

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Abstract

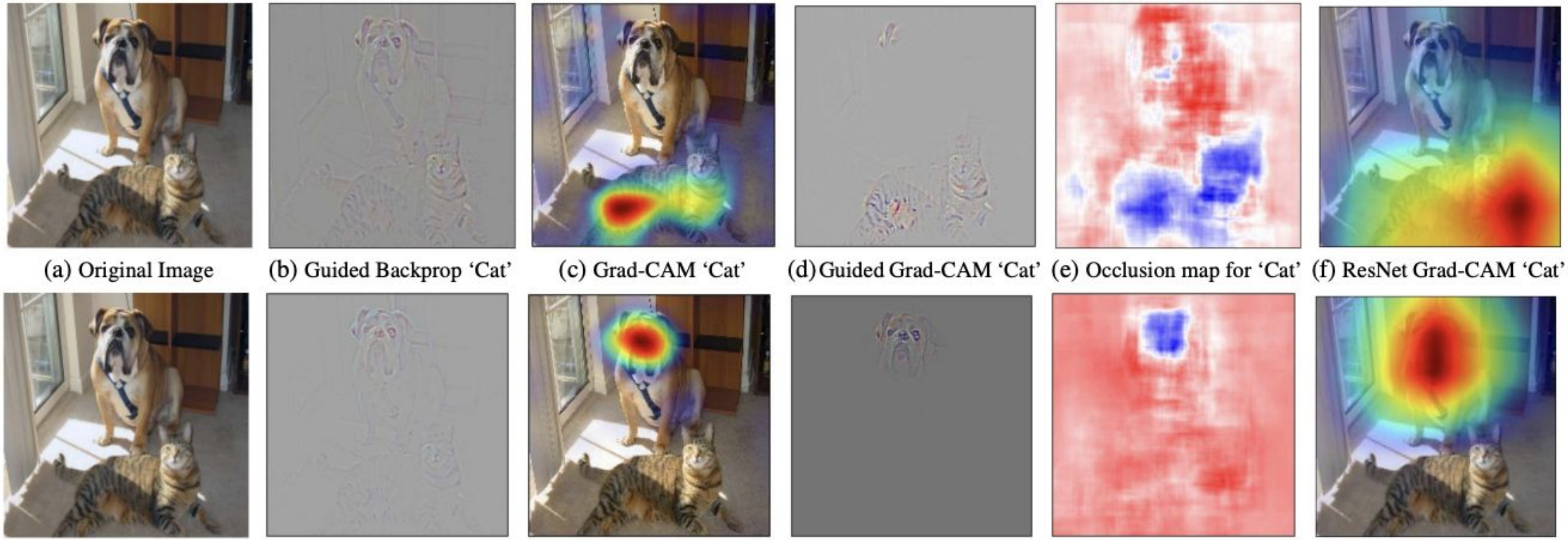
We propose a technique for producing 'visual explanations' for decisions from a large class of Convolutional Neural Network (CNN)-based models, making them more transparent. Our approach – Gradient-weighted Class Activation Mapping (Grad-CAM), uses the gradients of any target concept (say logits for 'dog' or even a caption), flowing into the final convolutional layer to produce a coarse localization map highlighting the important regions in the image for predicting the concept. Unlike previous approaches, Grad-CAM is applicable to a wide variety of CNN model-families: (1) CNNs with fully-connected layers (e.g. *VGG), (2) CNNs used for structured outputs* (e.g. *captioning), (3) CNNs used in tasks with multi-modal inputs* (e.g. *visual question an-*

1. Introduction

Convolutional Neural Networks (CNNs) and other deep networks have enabled unprecedented breakthroughs in a variety of computer vision tasks, from image classification [24, 16] to object detection [15], semantic segmentation [27], image captioning [43, 6, 12, 21], and more recently, visual question answering [3, 14, 32, 36]. While these deep neural networks enable superior performance, their lack of decomposability into *intuitive and understandable* components makes them hard to interpret [26]. Consequently, when today's intelligent systems fail, they fail spectacularly disgracefully, without warning or explanation, leaving a user staring at an incoherent output, wondering why.

Grad-Cam (Gradiente de Class Activation Mapping)

- Se utiliza para entender qué partes de una imagen contribuyen más a la clasificación realizada por una red neuronal convolucional.
- La técnica se basa en los gradientes del score de clasificación con respecto al feature map de convolución final de la red.
    - Identifica las regiones de la imagen que tienen un impacto significativo en la decisión de clasificación.
    - Las áreas con gradientes más grandes son aquellas en las que la puntuación final de la clasificación depende en mayor medida de los datos.



(a) Original Image (b) Guided Backprop 'Cat' (c) Grad-CAM 'Cat' (d) Guided Grad-CAM 'Cat' (e) Occlusion map for 'Cat' (f) ResNet Grad-CAM 'Cat'

(g) Original Image (h) Guided Backprop 'Dog' (i) Grad-CAM 'Dog' (j) Guided Grad-CAM 'Dog' (k) Occlusion map for 'Dog' (l) ResNet Grad-CAM 'Dog'

Figure 1: (a) Original image with a cat and a dog. (b-f) Support for the cat category according to various visualizations for VGG-16 and ResNet. (b) Guided Backpropagation [42]: highlights all contributing features. (c, f) Grad-CAM (Ours): localizes class-discriminative regions, (d) Combining (b) and (c) gives Guided Grad-CAM, which gives high-resolution class-discriminative visualizations. Interestingly, the localizations achieved by our Grad-CAM technique, (c) are very similar to results from occlusion sensitivity (e), while being orders of magnitude cheaper to compute. (f, l) are Grad-CAM visualizations for ResNet-18 layer. Note that in (c, f, i, l), red regions corresponds to high score for class, while in (e, k), blue corresponds to evidence for the class. Figure best viewed in color.

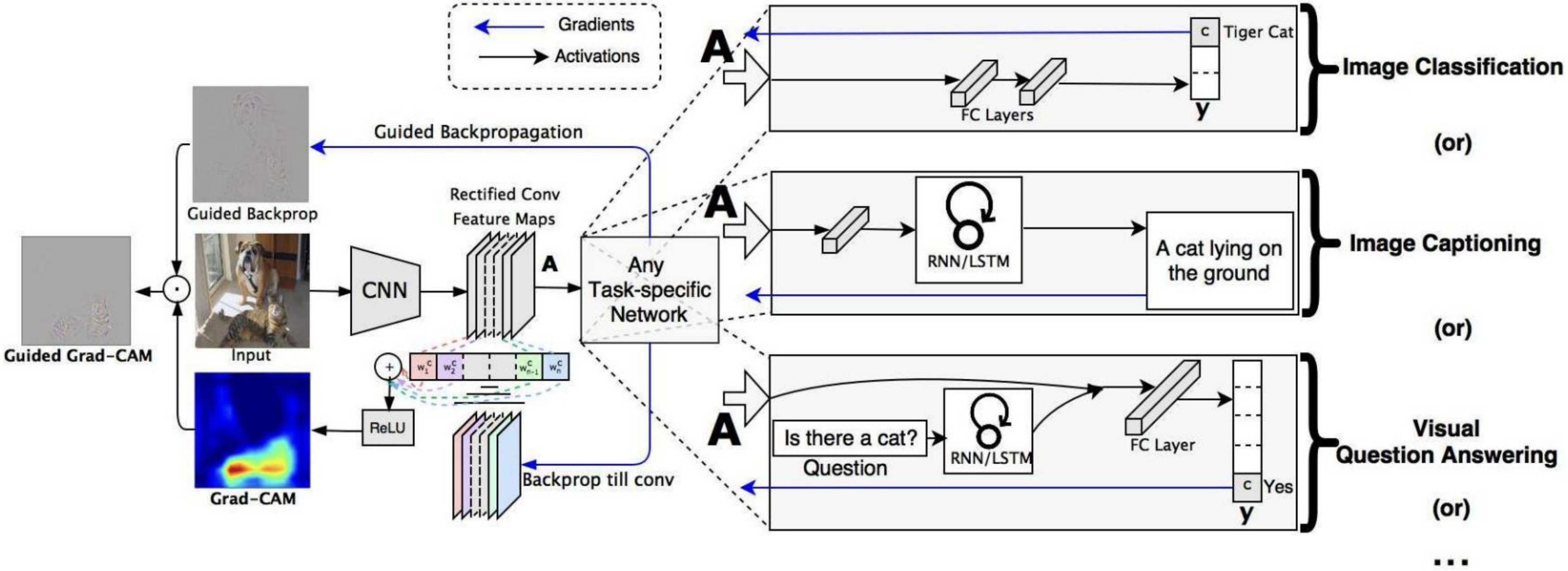
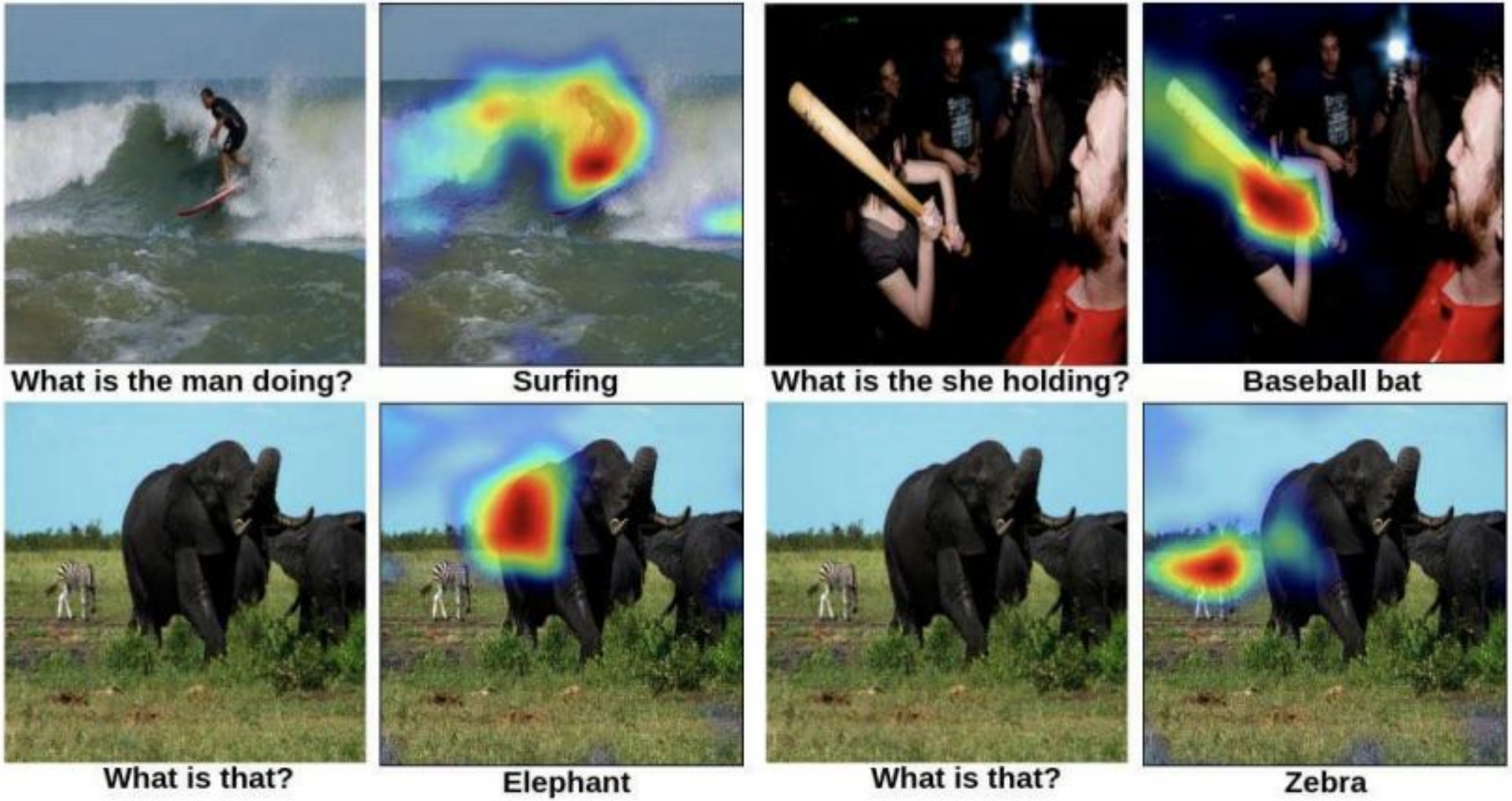


Figure 2: Grad-CAM overview: Given an image and a class of interest (*e.g.*, 'tiger cat' or any other type of differentiable output) as input, we forward propagate the image through the CNN part of the model and then through task-specific computations to obtain a raw score for the category. The gradients are set to zero for all classes except the desired class (tiger cat), which is set to 1. This signal is then backpropagated to the rectified convolutional feature maps of interest, which we combine to compute the coarse Grad-CAM localization (blue heatmap) which represents where the model has to look to make the particular decision. Finally, we pointwise multiply the heatmap with guided backpropagation to get Guided Grad-CAM visualizations which are both high-resolution and concept-specific.



(b) Visualizing ResNet based Hierarchical co-attention VQA model from [29]

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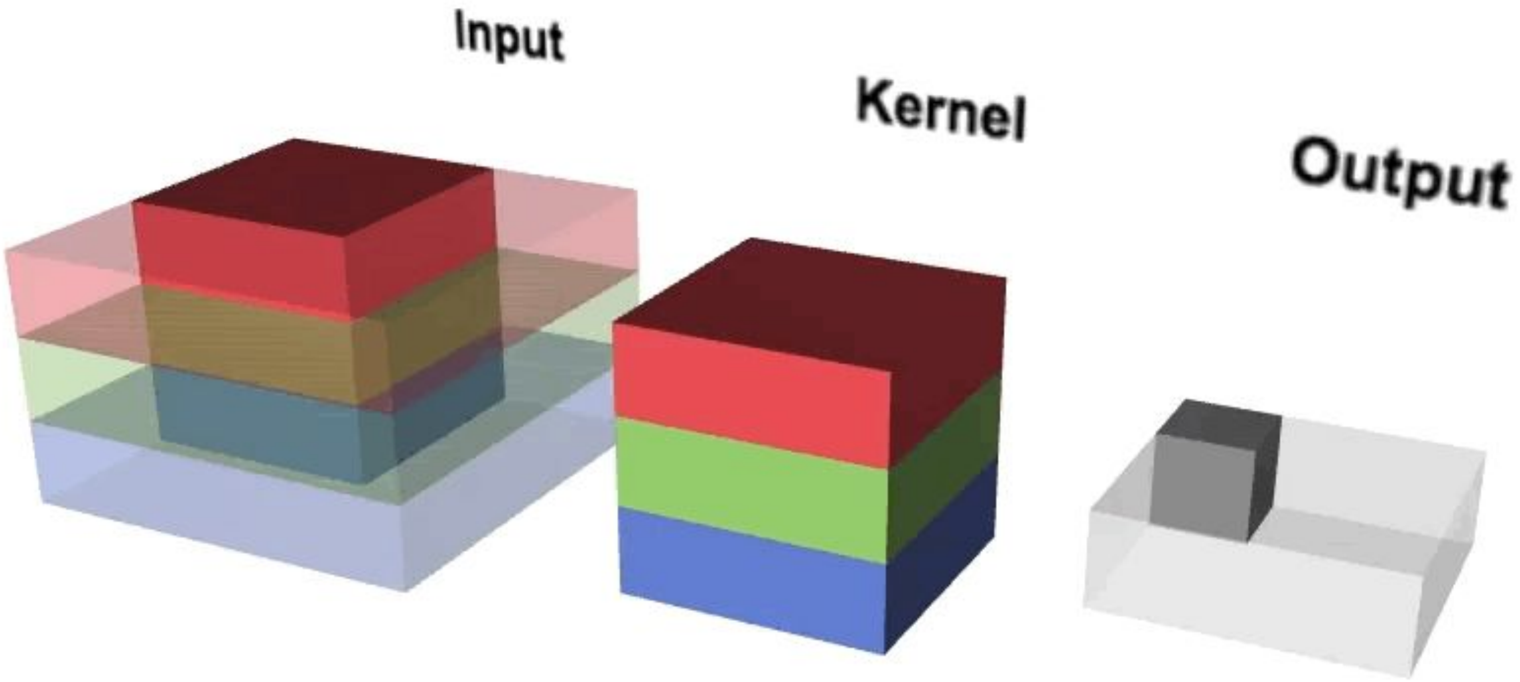
- ResNet: https://arxiv.org/pdf/1512.03385.pdf
- ZF net: https://arxiv.org/pdf/1311.2901.pdf
- Cam: https://openaccess.thecvf.com/content_cvpr_2015/papers/Oquab_Is_Object_Localization_2015_CVPR_paper.pdf
- Grad-cam: https://openaccess.thecvf.com/content_ICCV_2017/papers/Selvaraju_Grad-CAM_Visual_Explanations_ICCV_2017_paper.pdf


Input
Kernel
Output